

1 Mechanistic Explainability in Reinforcement
2 Learning

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The Golem Myth

A creation that surpasses its creator.

Long before modern technology, humanity had already questioned the possibility of losing control over its creations. One of the earliest expressions of this idea can be found in the story of the Golem. In Hebrew tradition, the term Golem refers to an artificial creature created by humans [Nocks, 1998]. These creatures are made of clay and brought to life using instructions inscribed on their bodies, as illustrated in Figure 1. The earliest accounts describe the Golem as a protective and helpful figure.

This vision becomes more ambiguous in the seventeenth century, as illustrated by the legend of the Golem of Prague. In this story, the initially small Golem gradually grows in size and strength until it can no longer be safely controlled [Nocks, 1998]. The creation intended to protect others therefore becomes a potential threat to them. The only remaining solution is to destroy the creature in order to protect the community.

The Golem has inspired similar narratives over time, including the modern notion of the robot. The concept of the robot as an artificial servant was introduced in the play *Rossum's Universal Robots*, written in 1920 [Čapek, 1920]. In the play, a scientist creates artificial beings designed to perform labour. In this first version, robots are organic creatures forged through advances in chemistry. Conceived as servants, these artificial beings turn against humanity and destroy civilisation.

The connection between the Golem and the robot was explicitly made by the author, as illustrated by his statement that “the Robot is the Golem made flesh through mass production” [Comrada, 1995]. Despite their destructive role in the play, the word robot is now used to designate artificial beings created to serve humans. This shows that, from its origin, the word robot has embodied the hopes and fears surrounding technological progress.

Today, this tension is particularly evident in the field of artificial intelligence. With the rise of computer technology, organic robots were gradually replaced in the collective imagination by mechanical ones, but the underlying fears remained. These concerns are even stronger as progress in the field has continued to accelerate. In these times of technological revolution, no one can predict how this development will unfold. What is certain, however, is that this field offers an almost literal adaptation of the Golem myth.

Among the subfields of artificial intelligence, reinforcement learning is most closely aligned with the goal of creating autonomous agents while minimizing reliance on human guidance. In this learning paradigm, an agent interacts with an environment, selects actions, and receives feedback according to the outcomes



Figure 1: The creation of a Golem, by Mikoláš Aleš [Aleš and Wenig, 1908].

44 of its behaviour [Sutton and Barto, 2018]. Through repeated interaction with its
 45 environment, the agent learns a strategy to achieve the objective. As a result, the
 46 agent's behavior is not explicitly programmed but learned through interactions
 47 with the environment [Amodei et al., 2016]. This minimal guidance enables
 48 reinforcement learning systems to achieve superhuman capabilities [Silver et al.,
 49 2017, OpenAI et al., 2019].

50 However, maintaining control over such agents presents a major challenge,
 51 as the autonomy that makes this learning approach powerful also makes their
 52 behavior difficult to control. Despite the difficulty, understanding the behaviour of
 53 such agents is the objective of this manuscript. This work therefore places within
 54 a long-standing line of thought that can be summarized by the following question:
 55 how can we ensure that our creations remain aligned with our intentions?

Chapter 1

Introduction

This initial chapter provides a general introduction to this work. It begins by presenting the research context that motivates the study. It then introduces the research questions addressed throughout the manuscript. To address these questions, it then presents the main contributions. Finally, it concludes with an overview of the manuscript's structure.

1.1 Research Context

As is evident in recent years, Artificial Intelligence (AI) has advanced rapidly.

This progress is driven largely, by major development in Deep Learning (DL). This type of approach relies on artificial Neural Networks (NNs), which offer an unprecedented level of performance. Inspired by their biological counterparts, these networks are known as connectionist models. In such models, information is represented and transformed through patterns of interactions among interconnected elementary computational units [Rumelhart et al., 1986].

One of the fields significantly transformed by DL is Reinforcement Learning (RL), which is the main focus of this manuscript. The purpose of RL is to enable an agent to learn how to make decisions through interactions with an environment [Sutton and Barto, 2018]. At each step, the agent observes the current state of the environment and selects an action. This action changes the state of the environment and results in a reward. The objective of the agent is then to learn a behavior that maximizes the cumulative rewards obtained over time. Due to their flexibility, NNs provide a powerful means of representing the agent's information-processing capabilities. Their use has consequently made it possible to apply RL to problems that were previously intractable [Mnih et al., 2013, Silver et al., 2017, OpenAI et al., 2019].

The ability of RL to operate in complex environments introduces a new challenge. When an agent learns to act in such environments, it can develop complex strategies. Although such agents can achieve strong performance, these strategies can be difficult to understand. From an external perspective, we observe sequences of interactions between the agent and its environment, but these observations are not enough to infer the agent's strategy.

An alternative approach is to analyze the agent itself. However, due to their connectionist nature, NN that model the agent's policy are not directly

interpretable. The information behind a decision is distributed across many units and connections making it impossible to trace. NNs are therefore often regarded as "black box" models [Guidotti et al., 2018].

To address this black-box problem, the field of eXplainable Artificial Intelligence (XAI) has emerged [Gunning and Aha, 2019, Barredo Arrieta et al., 2020]. In the specific context of RL, this research area is referred to as eXplainable Reinforcement Learning (XRL) [Puiutta and Veith, 2020, Milani et al., 2024]. XRL develops methods for producing explanations that improve our understanding of agent behavior. Such explanations can serve several purposes, including increasing human trust, identifying biases and verifying that previously identified problems have been effectively corrected [Barredo Arrieta et al., 2020, Milani et al., 2024].

This manuscript presents XRL methods from different approaches to provide a broader view of the field. We focus on the mechanistic approach to explainability, which we refer to as Mechanistic Explainability (ME) (Definition 1.1.1). Although still relatively underexplored, this approach seems particularly promising to us. In ME, the goal is to understand how information is represented within the NN. From this perspective, explainability is not achieved despite the use of NNs, but through them.

Definition 1.1.1: Mechanistic Explainability

ME is the approach that seeks to explain the behavior of an NN by analyzing its internal mechanisms.

To understand how NNs work internally, ME studies the intermediate representations learned by these networks. These representations are called Latent Representations (LRs) [Bengio et al., 2014]. Together, they form the network's Latent Space (LS). Analyzing the structure of this LS provides insights into how information relevant to decision-making is organized.

To develop an XRL method based on ME, we need to address several research questions, which are detailed in the following section.

1.2 Research Questions

The first question concerns the analysis of LRs themselves. Numerous methods have been proposed to analyze the LSs. These methods originate from different research communities are rarely compared within a common framework. This observation leads to our first research question:

Research Question 1

What approaches have been proposed in the literature to analyze the internal workings of NNs?

The second question concerns the organization of existing research in XRL. Several taxonomies have been proposed to organize existing XRL methods [Puiutta and Veith, 2020, Milani et al., 2024]. However, these taxonomies often rely on approaches initially developed for explainability in supervised settings. As a result, the organization of existing XRL methods remains ambiguous, making

128 it difficult to structure the field. This observation leads to our second research
 129 question:

Research Question 2

How can existing XRL techniques be organized into a taxonomy that accounts for the specificities of RL?

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131 The analysis of existing XRL approaches also reveals a third limitation.
 132 Among existing methods for explaining RL agents, few adopt a mechanistic
 133 perspective. This observation leads to our third research question:

Research Question 3

How can the internal representations of NNs be exploited to identify the mechanisms underlying the decision-making process of RL agents?

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135 The analysis of existing XRL methods highlights a fourth question concerning
 136 the evaluation of explanations. Although numerous approaches have been
 137 proposed to explain the behavior of RL agents, there is no widely accepted set of
 138 objective criteria for evaluating the quality and relevance of the explanations they
 139 produce. Existing evaluations often rely on user studies or qualitative analyses,
 140 making it difficult to systematically assess the validity of an explanation approach
 141 [Doshi-Velez and Kim, 2017, Milani et al., 2024].

142 This issue is particularly important for approaches that seek to explain agent
 143 behavior by analyzing LRs. Multiple hypotheses can be proposed about how
 144 information is structured within a LS. It is therefore important to define objective
 145 criteria for comparing these competing hypotheses. This leads to our fourth
 146 research question:

Research Question 4

How can the quality and relevance of explanations produced for RL agents be objectively quantified?

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1.3 Contributions

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149 To address these four research questions, this manuscript is based on three main
 150 contributions.

151 The first contribution has not yet been published and addresses Research
 152 Question 1. In this contribution, we make explicit the working hypothesis that
 153 underlies much of the literature on LS analysis. We then organize existing LS
 154 analysis methods into families to provide an overview of the different approaches.

155 The second contribution addresses Research Question 2. This contribution
 156 takes the form of a first paper [Alaarabiou et al., 2024] and an extended English
 157 version [Alaarabiou et al., 2026b], both of which were submitted for review and
 158 accepted for publication. In this contribution, we first clarify what explainability
 159 means in RL by providing a clear frame of reference. Building on these frame,
 160 we then conduct a comprehensive analysis of the state of the art in XRL.

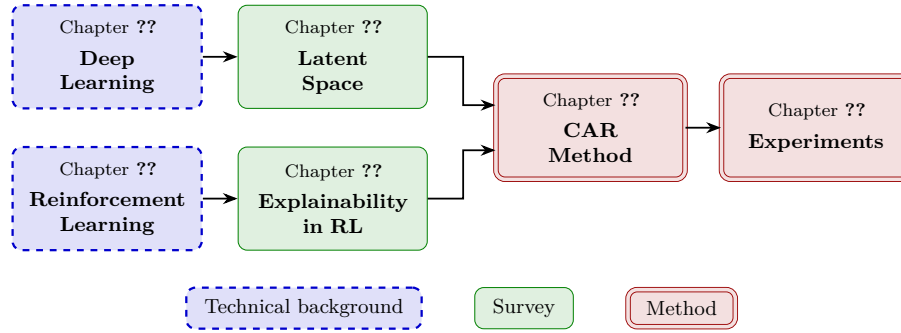


Figure 1.1: Structure of the manuscript.

The third contribution addresses Research Questions 3 and 4. In the paper [Alaarabiou et al., 2026a], which has been reviewed and accepted for publication, we introduce Clustering Agent Representation (CAR), an XRL method based on the principles of ME. By analyzing the LRs, CAR produces explanations in the form of clusters of states that are similar from the agent’s own perspective. We further show that this representation remains stable across training, indicating that the agent develops an objective representation of a given task.

This manuscript can be viewed as a synthesis of these three contributions.

1.4 Manuscript Structure

This manuscript brings together the contributions developed throughout the thesis. It is therefore written with the reviewers in mind while remaining as accessible as possible to readers. To this end, the background chapters introduce the concepts on which the subsequent chapters rely. Allowing readers to follow the manuscript without having to consult external material.

The manuscript is organized into a sequence of chapters, with each chapter falling into one of these three categories:

Technical background chapters introduce fundamental concepts required to understand the subsequent contributions.

Survey chapters review the existing literature around a specific research question. The organization adopted in each of these surveys are original and therefore constitute contributions of this manuscript.

Method chapters present CAR, our ME method for RL.

Following these categories, the chapters are organized as shown in Figure 1.1 and detailed below:

Deep Learning (Chapter ??, technical background)

This chapter introduces the fundamental concepts of DL.

Latent Space Analysis (Chapter ??, survey)

This chapter addresses Research Question 1 by providing a survey of existing approaches for analyzing the LSs.

190 **Reinforcement Learning (Chapter ??, technical background)**

191 This chapter introduces the fundamental concepts of RL.

192 **Explainability in Reinforcement Learning (Chapter ??, survey)**

193 This chapter addresses Research Question 2 by reviewing existing approaches
194 to explainability in RL.

195 **The CAR Method (Chapter ??, method)**

196 This chapter addresses Research Questions 3 and 4 by proposing a new
197 method for explainability in RL based on the analysis of LRs.

198 **Evaluation of the CAR Method (Chapter ??, method)**

199 This chapter presents the implementation and experimental evaluation of the
200 proposed method.

201 **Conclusion**

202 This chapter summarizes the main contributions of the manuscript, discusses
203 the limitations and presents possible directions for future research.

204 Now that the research context and objectives have been presented, we can
205 turn to the first chapter, which introduces the fundamentals of DL.

206 Acronyms

207 **AI** Artificial Intelligence 3

208 **CAR** Clustering Agent Representation 6

209 **DL** Deep Learning 3, 6, 7

210 **LR** Latent Representation 4–7

211 **LS** Latent Space 4–6

212 **ME** Mechanistic Explainability 4, 6

213 **NN** Neural Network 3–5

214 **RL** Reinforcement Learning 3–7

215 **XAI** eXplainable Artificial Intelligence 4

216 **XRL** eXplainable Reinforcement Learning 4–6

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